

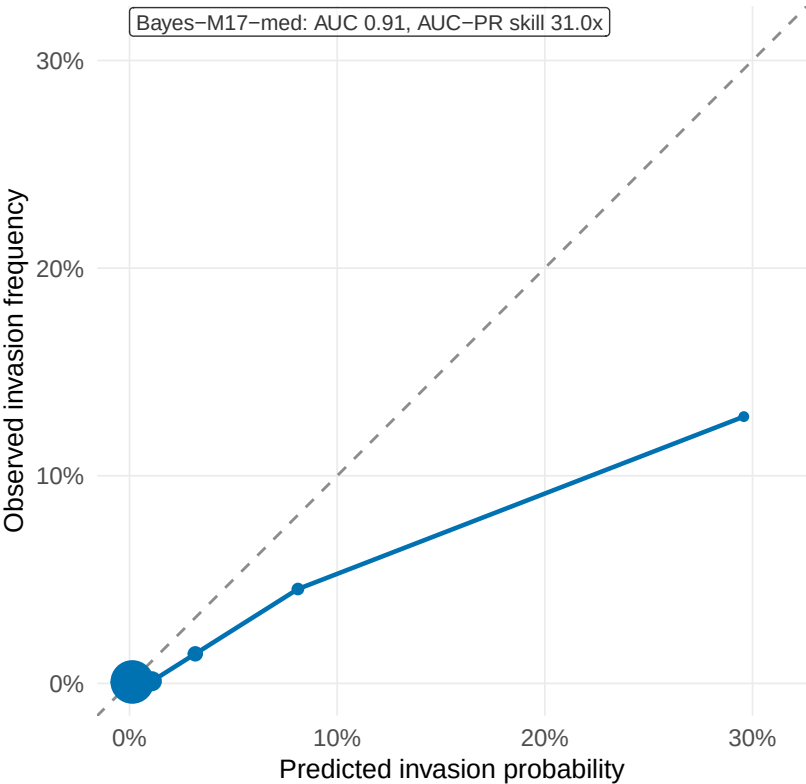
# How good is the invasion model? (h = 1w)

(A) calibration + AUC and (B) real-time prioritisation vs random – after Kraemer & Cauchemez 2017, Lancet Inf Dis, Fig 3

## (A) Calibration: predicted vs observed invasion (h = 1w)

On the diagonal = calibrated probabilities; below = over-confident.

Bayes-M17-med      zone-weeks      ● 1000      ● 2000      ● 3000      ● 4000      ● 5000

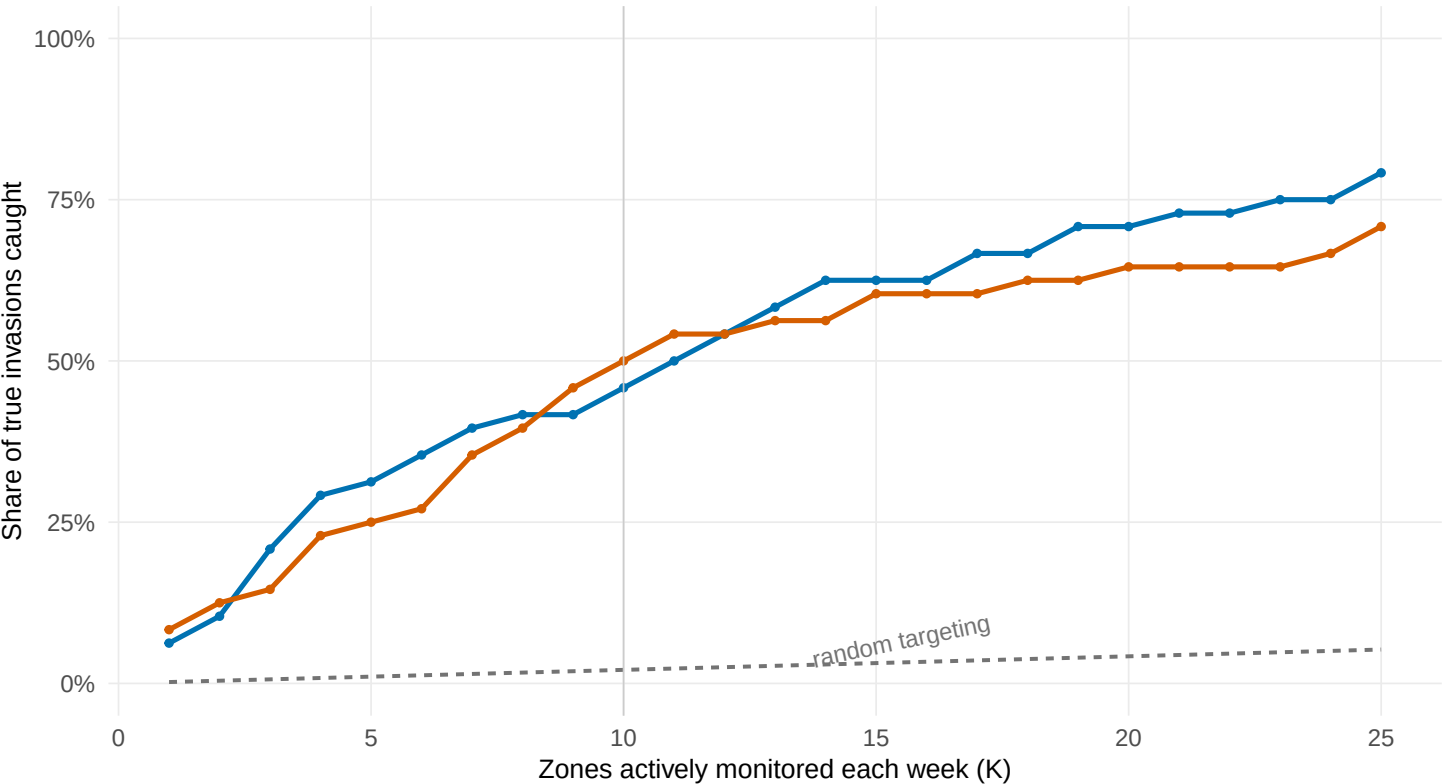


The 2.3% of at-risk zone-weeks we score >15%/week had a 13% observed invasion rate.

## (B) Prioritisation: invasions caught vs random targeting

Share of true next-week invasions caught at a fixed weekly alert budget of K zones, pooled over LFO folds

Bayes-M17-med      Naive-epicentre-inflow



At a budget of K=10 zones/week, Naive-epicentre-inflow catches 50% of the next invasions vs 2% for random targeting (23.8x b